



B.Tech. Degree III Semester Examination November 2018

CS 15-1306 DATA AND COMPUTER COMMUNICATION

(2015 Scheme)

Time : 3 Hours

Maximum Marks : 60

PART A

(Answer **ALL** questions)

(10 × 2 = 20)

- I. (a) A nonperiodic composite signal contains frequencies from 10 to 30 KHz. The peak amplitude is 10V for the lowest and the highest signals, and is 30V for the 20 KHz signals. Assuming that the amplitude changes gradually from the minimum to the maximum. Draw the frequency spectrum and calculate bandwidth.
- (b) What is a quantization error?
- (c) Eight channels each with a 100 KHz bandwidth are to be multiplexed together. What is the minimum bandwidth of the link if there is a need for guard band of 10 KHz between channels to prevent interference?
- (d) Explain about direct sequence spread spectrum.
- (e) Explain about multiplexing process.
- (f) What is a LOS propagation?
- (g) Differentiate the term 'error detection' and 'error correction'.
- (h) What is a codeword? Explain with one example.
- (i) What are the different properties of an Ethernet?
- (j) What are passive and active hubs?

PART B

(4 × 10 = 40)

- II. (a) Encode the data 01010101 by NRZ-L, NRZ-I, Manchester and MLT3, assuming that the last signal level has been positive. (8)
- (b) Explain the duties of Physical layer. (2)
- OR**
- III. Explain in detail about transmission impairments. (10)
- IV. Discuss about guided transmission media. (10)
- OR**
- V. Explain the following: (10)
 - (i) Circuit Switched Networks
 - (ii) Datagram Networks
- VI (a) With the help of diagram explain go back N ARQ. (7)
- (b) Explain about hamming code with examples. (3)
- OR**
- VII. A message block 11100101 transmitted across a data link using a CRC for error detection. A generator polynomial of 11011 is to be used. (10)
 - (i) Show the generation of the codeword at the sender site.
 - (ii) Show the checking of the codeword at the receiver site (received data was 111001010111).
- VIII. (a) What are the categories of Fast Ethernet? (4)
- (b) Explain the function of (i) repeater (ii) router (iii) gateway. (6)
- OR**
- IX. (a) With the help of diagram explain about ring and star topologies. (5)
- (b) Explain about piconet and scatternet. (5)